



Configuring RSTP and MSTP to Improve STP Performance

Difficulty (out of 4 stars): ★★★★★

Recommended study time: 3 hours

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1 About This Lab

1.1 Task Description

RSTP makes improvements over STP to implement fast network topology convergence. However, when RSTP or STP is running on a network that is divided into VLANs, all VLANs on the LAN share one spanning tree. The blocked links do not transmit any traffic, and load balancing cannot be implemented among VLANs. As a result, the link bandwidth utilization and device resource utilization are low.

To overcome the defects of RSTP and STP, the IEEE released the 802.1s standard in 2002, which defined the Multiple Spanning Tree Protocol (MSTP). MSTP is compatible with STP and RSTP. It sets up multiple loop-free trees to prevent broadcast storms and implement redundancy.

After completing this task, you will be able to configure Huawei switches, understand RSTP and MSTP, and know how to use RSTP and MSTP to optimize a switched network.

1.2 Objectives

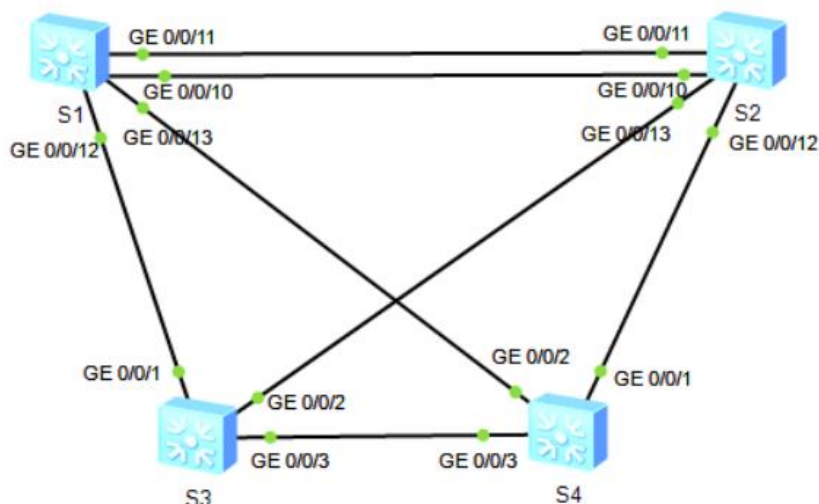
On completing this task, you will be able to:

1. Manually change a bridge priority to affect root bridge election.
2. Manually change a port cost to control root port election.
3. Manually change a port priority value to control root port election.
4. Configure MSTP to implement load balancing among VLANs.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology



Redundancy is deployed on a company's network, and RSTP is configured to prevent loops. All VLANs share one STP spanning tree. To implement load balancing of data traffic between VLANs, MSTP is deployed on the network.



2.2 Initial Configuration

- Initial configuration of S1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S1
[S1]
```

- Initial configuration of S2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S2
[S2]
```

- Initial configuration of S3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S3
[S3]
```

- Initial configuration of S4

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S4
[S4]
```

3 Task Implementation

3.1 Checking Basic Configurations

The details are not provided.

3.2 Configuring Basic RSTP Functions

1. Enable STP on S1, S2, S3, and S4, and switch the STP mode to RSTP.

- Configuration on S1

```
[S1]stp enable
[S1]stp mode rstp
```

- Configuration on S2

```
[S2]stp enable
[S2]stp mode rstp
```

- Configuration on S3

```
[S3]stp enable
[S3]stp mode rstp
```

- Configuration on S4

```
[S4]stp enable
[S4]stp mode rstp
```



2. Check the STP status and brief statistics information.

```
<S1>display stp instance 0 brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/10	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/11	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/12	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/13	DESI	FORWARDING	NONE

```
[S2]display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/10	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/11	ALTE	DISCARDING	NONE
0	GigabitEthernet0/0/12	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/13	DESI	FORWARDING	NONE

```
[S3]display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/1	ALTE	DISCARDING	NONE
0	GigabitEthernet0/0/2	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/3	ALTE	DISCARDING	NONE

```
[S3]
```

```
[S4]display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/1	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/2	ALTE	DISCARDING	NONE
0	GigabitEthernet0/0/3	DESI	FORWARDING	NONE

```
[S4]
```

All ports on S1 are designated ports, and S1 is the root bridge. The actual experimental result may be different from the preceding result due to the uncertainty of the switch MAC address.

3. Check the STP status and statistics on S1. The following information is displayed:

```
<S1>display stp
```

```
-----[CIST Global Info][Mode RSTP]-----
```

```

CIST Bridge           :32768.4c1f-cc1d-61a8
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC        :32768.4c1f-cc1d-61a8 / 0
CIST RegRoot/IRPC     :32768.4c1f-cc1d-61a8 / 0
CIST RootPortId       :0.0
BPDU-Protection       :Disabled
TC or TCN received    :15
TC count per hello    :0
STP Converge Mode     :Normal
Time since last TC    :0 days 0h:11m:14s
Number of TC          :17
Last TC occurred      :GigabitEthernet0/0/13

```

S1 is the root bridge.



3.3 Controlling Root Bridge Election

Configure S1 as the primary root bridge and S2 as the secondary root bridge.

1. Manually adjust the STP priority to specify S1 as the primary root bridge and S2 as the secondary root bridge.

```
[S1]stp priority 4096
```

```
[S2]stp priority 8192
```

When the default bridge priority (32768) is retained on the other two switches, S1 has the smallest bridge priority and S2 has the second smallest bridge priority.

2. Check the STP status and statistics on S1. The following information is displayed:

```
[S1]display stp
-----[CIST Global Info][Mode RSTP]-----
CIST Bridge           :4096 .4c1f-cc1d-61a8
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC        :4096 .4c1f-cc1d-61a8 / 0
CIST RegRoot/IRPC     :4096 .4c1f-cc1d-61a8 / 0
CIST RootPortId       :0.0
BPDU-Protection       :Disabled
TC or TCN received    :75
TC count per hello    :0
STP Converge Mode     :Normal
Time since last TC    :0 days 0h:1m:16s
Number of TC          :45
Last TC occurred      :GigabitEthernet0/0/10
```

S1 has the bridge priority of 4096 and is still the root bridge.

3. Delete the manually configured bridge priorities on S1 and S2, and run the **stp root** command to specify the root bridge and secondary root bridge.

```
[S1]undo stp priority
```

```
[S1]stp root primary
```

```
[S2]undo stp priority
```

```
[S2]stp root secondary
```

4. Check the STP status and statistics on S1 and S2. The following information is displayed:

```
[S1]display stp
-----[CIST Global Info][Mode RSTP]-----
CIST Bridge           :0 .4c1f-cc1d-61a8
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC        :0 .4c1f-cc1d-61a8 / 0
CIST RegRoot/IRPC     :0 .4c1f-cc1d-61a8 / 0
CIST RootPortId       :0.0
BPDU-Protection       :Disabled
CIST Root Type        :Primary root
TC or TCN received    :85
TC count per hello    :0
STP Converge Mode     :Normal
Time since last TC    :0 days 0h:0m:9s
```



```

Number of TC          :51
Last TC occurred      :GigabitEthernet0/0/10

[S2]display stp
-----[CIST Global Info][Mode RSTP]-----
CIST Bridge           :4096 .4c1f-cc69-5bf7
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC        :0 .4c1f-cc1d-61a8 / 20000
CIST RegRoot/IRPC     :4096 .4c1f-cc69-5bf7 / 0
CIST RootPortId       :128.10
BPDU-Protection       :Disabled
CIST Root Type        :Secondary root
TC or TCN received    :213
TC count per hello    :0
STP Converge Mode     :Normal
Time since last TC    :0 days 0h:0m:35s
Number of TC          :44
Last TC occurred      :GigabitEthernet0/0/12
  
```

The bridge priority of S1 is 0, and that of S2 is 4096. S1 is the root bridge, and S2 is the secondary root bridge.

3.4 Modifying the Interface Cost to Control Root Port Election

1. Check the STP status and brief statistics on S4.

```
[S4]display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/1	ALTE	DISCARDING	NONE
0	GigabitEthernet0/0/2	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/3	DESI	FORWARDING	NONE

GE0/0/2 on S4 has a smaller RPC and becomes the root port.

2. Check the STP status and statistics on GE0/0/2 of S4.

```
[S4]display stp interface GigabitEthernet 0/0/2
```

```

-----[CIST Global Info][Mode RSTP]-----
CIST Bridge           :32768.4c1f-cc49-4c7c
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC        :0 .4c1f-cc1d-61a8 / 20000
CIST RegRoot/IRPC     :32768.4c1f-cc49-4c7c / 0
CIST RootPortId       :128.2
BPDU-Protection       :Disabled
TC or TCN received    :98
TC count per hello    :0
STP Converge Mode     :Normal
Time since last TC    :0 days 0h:8m:35s
Number of TC          :47
Last TC occurred      :GigabitEthernet0/0/2
----[Port2(GigabitEthernet0/0/2)][FORWARDING]----
  
```



```

Port Protocol      :Enabled
Port Role          :Root Port
Port Priority       :128
Port Cost(Dot1T )  :Config=auto / Active=20000
Designated Bridge/Port :0.4c1f-cc1d-61a8 / 128.13
Port Edged         :Config=default / Active=disabled
Point-to-point     :Config=auto / Active=true
Transit Limit      :147 packets/hello-time
Protection Type    :None
Port STP Mode      :RSTP
Port Protocol Type :Config=auto / Active=dot1s
BPDU Encapsulation :Config=stp / Active=stp
PortTimes          :Hello 2s MaxAge 20s FwDly 15s RemHop 0
TC or TCN send     :26
TC or TCN received :40
BPDU Sent          :1747
                   TCN: 0, Config: 0, RST: 1747, MST: 0
BPDU Received      :1048
                   TCN: 0, Config: 0, RST: 1048, MST: 0

```

At this time, the path cost calculation standard is dot1t, and the STP cost of the port is 20000.

3. Change the STP cost of GE0/0/2 on S4 to 40001.

```

[S4]interface GigabitEthernet 0/0/2
[S4-GigabitEthernet0/0/2] stp cost 40001

```

4. Check the STP status and brief statistics on S4 again.

```
<S4>display stp brief
```

MSTID Port	Role STP State	Protection
0 GigabitEthernet0/0/1	ROOT FORWARDING	NONE
0 GigabitEthernet0/0/2	ALTE DISCARDING	NONE
0 GigabitEthernet0/0/3	ALTE DISCARDING	NONE

The RPC of GE0/0/1 is 40000, which is smaller than the RPC of GE0/0/2 (40001). Therefore, GE0/0/1 on S4 becomes the root port.

3.5 Modifying the Interface Priority to Control Root Port Election

1. Check the STP status and brief statistics on S2.

```
[S2]display stp brief
```

MSTID Port	Role	STP State	Protection
0 GigabitEthernet0/0/10	ROOT	FORWARDING	NONE
0 GigabitEthernet0/0/11	ALTE	DISCARDING	NONE
0 GigabitEthernet0/0/12	DESI	FORWARDING	NONE
0 GigabitEthernet0/0/13	DESI	FORWARDING	NONE

On S2, GE0/0/10 and GE0/0/11 receive BPDUs with the same RPC, bridge ID, and interface priority. In this case, the interface IDs in the received BPDUs are compared.

2. Enable LLDP on S1 and S2 and check the interface interconnection relationship.



```
[S1]lldp enable
```

```
[S2]lldp enable
```

```
[S2]display lldp neighbor brief
```

Local Intf	Neighbor Dev	Neighbor Intf	Exptime
GE0/0/10	S1	GE0/0/10	102
GE0/0/11	S1	GE0/0/11	102
GE0/0/12	S4	GE0/0/1	108
GE0/0/13	S3	GE0/0/2	103

GE0/0/10 of S2 is connected to GE0/0/10 of S1, and GE0/0/11 of S2 is connected to GE0/0/11 of S1. GE0/0/10 of S2 receives BPDUs with a smaller interface ID. Therefore, GE0/0/10 of S2 becomes the root port.

- On S1, change the STP interface priority of GE0/0/11 so that the BPDUs sent by GE0/0/11 are better than those sent by GE0/0/10.

```
[S1]interface GigabitEthernet 0/0/11
```

```
[S1-GigabitEthernet0/0/11] stp port priority 64
```

The STP interface priority value is 128. A smaller value indicates a higher priority.

- Check the STP status and brief statistics on S2 again.

GE0/0/1 of S2 becomes the root port.

```
[S2]display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/10	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/11	ALTE	DISCARDING	NONE
0	GigabitEthernet0/0/12	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/13	DESI	FORWARDING	NONE

3.6 Basic MSTP Configurations

Create VLANs 10, 20, 30, 40, 50, 60, 70, and 80 on all switches, configure the MSTP region **hcip**, and create two new instances: instance 1 and instance 2. Map VLANs 10, 30, 50, and 70 to instance 1, and map VLANs 20, 40, 60, and 80 to instance 2. Plan S1 as the primary root bridge of MSTI1 and the secondary root bridge of MSTI2, and plan S2 as the primary root bridge of MSTI2 and the secondary root bridge of MSTI1.

- Create VLANs.

```
[S1]vlan batch 10 20 30 40 50 60 70 80
```

```
[S2]vlan batch 10 20 30 40 50 60 70 80
```

```
[S3]vlan batch 10 20 30 40 50 60 70 80
```

```
[S4]vlan batch 10 20 30 40 50 60 70 80
```

- Configure the interconnection interfaces as trunk interfaces and allow packets from all VLANs to pass through.

The details are not provided.

- Change the working mode from STP to MSTP.



```
[S1]stp mode mstp
```

```
[S2]stp mode mstp
```

```
[S3]stp mode mstp
```

```
[S4]stp mode mstp
```

4. Configure MSTP.

```
[S1]stp region-configuration
```

```
[S1-mst-region] region-name hcip
```

```
[S1-mst-region] revision-level 1
```

```
[S1-mst-region] instance 1 vlan 10 30 50 70
```

```
[S1-mst-region] instance 2 vlan 20 40 60 80
```

```
[S1-mst-region] active region-configuration
```

```
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S1-mst-region] quit
```

```
[S2]stp region-configuration
```

```
[S2-mst-region] region-name hcip
```

```
[S2-mst-region] revision-level 1
```

```
[S2-mst-region] instance 1 vlan 10 30 50 70
```

```
[S2-mst-region] instance 2 vlan 20 40 60 80
```

```
[S2-mst-region] active region-configuration
```

```
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S2-mst-region] quit
```

```
[S3]stp region-configuration
```

```
[S3-mst-region] region-name hcip
```

```
[S3-mst-region] revision-level 1
```

```
[S3-mst-region] instance 1 vlan 10 30 50 70
```

```
[S3-mst-region] instance 2 vlan 20 40 60 80
```

```
[S3-mst-region] active region-configuration
```

```
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S3-mst-region] quit
```

```
[S4]stp region-configuration
```

```
[S4-mst-region] region-name hcip
```

```
[S4-mst-region] revision-level 1
```

```
[S4-mst-region] instance 1 vlan 10 30 50 70
```

```
[S4-mst-region] instance 2 vlan 20 40 60 80
```

```
[S4-mst-region] active region-configuration
```

```
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S4-mst-region] quit
```

5. Check the mappings between MSTP instances and VLANs on S1.

```
[S1]display stp region-configuration
```

```
Oper configuration
```

```
Format selector      :0
```

```
Region name          :hcip
```

```
Revision level       :1
```



Instance	VLANs Mapped
0	1 to 9, 11 to 19, 21 to 29, 31 to 39, 41 to 49, 51 to 59, 61 to 69, 71 to 79, 81 to 4094
1	10, 30, 50, 70
2	20, 40, 60, 80

6. Configure S1 as the primary root bridge of MSTI1 and the secondary root bridge of MSTI2.

```
[S1]stp instance 1 root primary
[S1]stp instance 2 root secondary
```

7. Configure S2 as the primary root bridge of MSTI2 and the secondary root bridge of MSTI1.

```
[S2]stp instance 1 root secondary
[S2]stp instance 2 root primary
```

8. Check the status and statistics of MSTI1 on S1.

```
[S1]display stp instance 1 brief
```

MSTID	Port	Role	STP State	Protection
1	GigabitEthernet0/0/10	DESI	FORWARDING	NONE
1	GigabitEthernet0/0/11	DESI	FORWARDING	NONE
1	GigabitEthernet0/0/12	DESI	FORWARDING	NONE
1	GigabitEthernet0/0/13	DESI	FORWARDING	NONE

The command output shows that all interfaces on S1 are designated interfaces, and S1 is the root bridge of MSTI1.

9. Check the status and brief statistics of MSTI2 on S2.

```
[S2]display stp instance 2 brief
```

MSTID	Port	Role	STP State	Protection
2	GigabitEthernet0/0/10	DESI	FORWARDING	NONE
2	GigabitEthernet0/0/11	DESI	FORWARDING	NONE
2	GigabitEthernet0/0/12	DESI	FORWARDING	NONE
2	GigabitEthernet0/0/13	DESI	FORWARDING	NONE

The command output shows that all interfaces on S2 are designated interfaces, and S2 is the root bridge of MSTI2.

4 Verification

Omitted.

5 Quiz

1. What optimizations does RSTP make to STP?
2. Compared with RSTP, what optimizations does MSTP make?
3. (True or false) The CIST is a tree consisting of the internal spanning tree and common spanning tree. (A)
 - A. True
 - B. False

